

**SECTION 11 3013
RESIDENTIAL APPLIANCES**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Kitchen appliances.

1.02 REFERENCE STANDARDS

- A. UL (DIR) - Online Certifications Directory; Current Edition.

1.03 SUBMITTALS

- A. See Section 01 3000 - Administrative Requirements, for submittal procedures.
- B. Product Data: Manufacturer's data indicating dimensions, capacity, and operating features of each piece of residential equipment specified.
- C. Copies of Warranties: Submit manufacturer warranty and ensure that forms have been completed in Owner's name and registered with manufacturer.

1.04 QUALITY ASSURANCE

- A. Electric Appliances: Listed and labeled by UL (DIR) and complying with NEMA Standards (National Electrical Manufacturers Association).

1.05 WARRANTY

- A. See Section 01 7800 - Closeout Submittals, for additional warranty requirements.
- B. Provide ten (10) year manufacturer warranty on magnetron tube of microwave ovens.

PART 2 PRODUCTS

2.01 KITCHEN APPLIANCES

- A. Provide Equipment Eligible for Energy Star Rating: Energy Star Rated.
- B. Microwave, Type ____: Countertop.
 - 1. Capacity: 0.7 cubic ft (0.019 cu m).
 - 2. Power: 700 watts.
 - 3. Features: Include turntable and 2-speed exhaust fan.
 - 4. Exterior Finish: Black.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify utility rough-ins are provided and correctly located.

3.02 INSTALLATION

- A. Install in accordance with manufacturer's instructions.

3.03 ADJUSTING

- A. Adjust equipment to provide efficient operation.

3.04 CLEANING

- A. Remove packing materials from equipment and properly discard.
- B. Wash and clean equipment.

END OF SECTION

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**SECTION 11 4000
FOODSERVICE EQUIPMENT**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Foodservice equipment.
- B. Connections to utilities.

1.02 REFERENCE STANDARDS

- A. ASTM A653/A653M - Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process; 2023.
- B. ASTM A666/A666M - Standard Specification for Annealed or Cold-Worked Austenitic Stainless Steel Sheet, Strip, Plate, and Flat Bar; 2024.
- C. ASTM C1036 - Standard Specification for Flat Glass; 2021.
- D. UL (DIR) - Online Certifications Directory; Current Edition.

1.03 SUBMITTALS

- A. See Section 01 3000 - Administrative Requirements, for submittal procedures.
- B. Product Data: Provide data on appliances; indicate configuration, sizes, materials, finishes, locations, and utility service connection locations, service characteristics, and wiring diagrams.
- C. Operation Data: Provide operating data for the specified equipment.
- D. Maintenance Data: Provide lubrication and periodic maintenance requirement schedules.
- E. Warranty: Submit manufacturer warranty and ensure forms have been completed in Owner's name and registered with manufacturer.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Store products clear of floor in a manner to prevent damage.
- B. Coordinate size of access and route to place of installation.

1.05 WARRANTY

- A. See Section 01 7800 - Closeout Submittals, for additional warranty requirements.
- B. Correct defective work of this section within a five year period after Date of Substantial Completion.
- C. Provide five year manufacturer warranty for replacement or repair of scheduled equipment, refrigerant and compressors, including disconnection and removal of defective unit, and connection of replacement unit.

PART 2 PRODUCTS

2.01 MANUFACTURERS

- A. Foodservice Equipment:
 - 1. Hobart Corp; ____: www.hobartcorp.com/#sle.
 - 2. Vulcan Hart Corp; ____: www.vulcanhart.com/#sle.
 - 3. Advance Tabco; <https://advancetabco.com/>.
 - 4. True Manufacturing; <https://www.truemfg.com/>
 - 5. Substitutions: See Section 01 6000 - Product Requirements.

2.02 REGULATORY REQUIREMENTS

- A. Comply with applicable codes for utility requirements.

2.03 EQUIPMENT

- A. Equipment Schedule: Refer to drawings.
 - 1. Equipment Eligible for Energy Star Rating: Provide Energy Star Rated equipment

2. Cooler and Freezer Units: Listed by UL (DIR).
 3. Electrical Wiring and Components and Self-Contained Refrigeration Systems: Comply with UL (DIR) listed product standards.
- B. Installation Accessories: Provide rough-in hardware, supports and connections, attachment devices, closure trim, and accessories as required for complete installation.

2.04 MATERIALS

- A. Sheet Steel: Hot-dipped galvanized steel sheet, ASTM A653/A653M, with G90/Z275 coating.
- B. Stainless Steel Sheet: ASTM A666/A666M Type 304 commercial grade, No. 4 finish.
- C. Glass: ASTM C1036 annealed, and laminated, 4 mm thick; exposed edges ground; cut or drilled to receive hardware.
- D. Finish Hardware: Manufacturer's standard.
- E. Service Outlet Covers and Escutcheons: Stainless Steel.

2.05 FINISHES

- A. Components: Shop finish.
- B. Stainless Steel: No. 4 finish.

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify ventilation outlets, service connections, and supports are correct and in required location.
- B. Verify that electric power is available and of the correct characteristics.

3.02 INSTALLATION

- A. Install items in accordance with manufacturers' instructions.
- B. Insulate to prevent electrolysis between dissimilar metals.

3.03 CLOSEOUT ACTIVITIES

- A. At completion of work, provide qualified and trained personnel to demonstrate operation of each item of equipment and instruct Owner in operating procedures and maintenance.
 1. Test equipment prior to demonstration.

3.04 PROTECTION

- A. Remove protective coverings from prefinished work.
- B. Protect finished work from damage.

END OF SECTION

**SECTION 11 6843
FOOTBALL/TRACK SCOREBOARD**

PART 1 GENERAL

1.01 SECTION INCLUDES

- A. Single-sided LED football/track scoreboard

1.02 REFERENCES

- A. Standard for Electric Signs, UL 48
- B. Standard for CSA C22.2 #207
- C. Federal Communications Commission Regulation Part 15
- D. National Electric Code

1.03 SUBMITTALS

- A. Product data: Submit manufacturer's product illustrations, data and literature that fully describe the scoreboards and accessories proposed for installation.
- B. Shop drawings: Submit mechanical and electrical drawings.
- C. Maintenance data: Submit manufacturer's installation, operation, and maintenance manuals.

1.04 DELIVERY, STORAGE, AND HANDLING

- A. Product delivered on site
- B. Scoreboard and equipment to be housed in a clean, dry environment

1.05 PROJECT CONDITIONS

- A. Environmental limitations: Do not install scoreboard equipment until mounting structure is secure and concrete has ample time to cure.
- B. Field measurements: Verify position and elevation of structure and its layout for scoreboard equipment. Verify dimensions by field measurements.
- C. Verify mounting structure is capable of supporting the scoreboard's weight and windload in addition to the auxiliary equipment.
- D. Installation may proceed within acceptable weather conditions.

1.06 QUALITY ASSURANCE

- A. For outdoor use
- B. Source Limitations: Obtain each type of scoring or related equipment through one source from a single manufacturer.
- C. ETL listed to UL 48
- D. NEC compliant
- E. FCC compliant
- F. ETLC listed to CSA 22.2 #207

1.07 WARRANTY

- A. Provide 5 years of no cost parts exchange including standard shipping on electronics parts and radios due to manufacturing defects
- B. Provide toll-free service coordination
- C. Provide technical online and phone support during Daktronics business hours

PART 2 PRODUCTS

2.01 MANUFACTURER

- A. Daktronics, Inc., 201 Daktronics Drive, P.O. Box 5128, Brookings, SD 57006-5128

- B. _____
- C. _____

2.02 PRODUCT

- A. Daktronics FB-2022 single-sided football/track scoreboard displays period time to 99:59, HOME and GUEST scores to 99, and DOWN/TO GO/BALL ON/QTR (quarter) information. Track running time, lead split and finish times display to 99:59.9. T.O.L. (time outs left) to nine are optional. Arrows indicate possession. Scoreboard comes standard with track captions on changeable panels. During the last minute of the period, the clock displays time to 1/10 of a second.

2.03 SCOREBOARD

- A. General information
1. Dimensions: 8'-0" (2.67 yard (2.44 meter)) high, 25'-0" (8.33 yard (7.62 meter)) wide, 0'-8" (7.99 inch (203 mm)) deep
 2. Base weight: 820 lb (372 kg) with vinyl captions – options may increase weight
 3. Base power requirement: 320 W (red/amber digits), 670 W (white digits) with vinyl captions – options may increase wattage
 4. Color: provide over 150 colors to choose from
- B. Construction
1. Alcoa aluminum alloy 5052 for excellent corrosion resistance
 2. Scoreboard back, face, and perimeter: 0.063" (0.06 inch (1.6 mm)) thick
 3. Scoreboard top and bottom: 0.125" (0.13 inch (3.18 mm)) thick
- C. Digits & Indicators
1. LED color:
 2. Red
 3. Clock digits: 30" (30 inch (762 mm)) high
 4. HOME, GUEST, DOWN, TO GO, BALL ON, and QTR digits: 24" (24.02 inch (610 mm)) high
 5. Seven bar segments per digit
 6. PanaView® LED digit technology
 7. All digits and indicators are sealed front and back with weather-tight silicone gel
- D. Captions
1. Vinyl applied directly to scoreboard face; track captions are on changeable panels
 2. HOME and GUEST captions: 15" (15 inch (381 mm)) high
 3. DOWN, TO GO, BALL ON, QTR, and track captions: 12" (12.01 inch (305 mm)) high
 4. Color: standard white or others available upon request
- E. Accessory Equipment
1. Vinyl striping applied around the clock and scoreboard face
 2. Custom team name caption in place of HOME
 3. Electronic Captions – add 160 lbs (73 kg) and 180 W (360 for white)
 4. Two 42" (31.73 inch (806 mm)) high, 31.75" (1.17 yard (1.07 meter)) wide vinyl logo/sponsor areas on lower corners of display face
 5. Soccer captions on changeable panels
 6. Lacrosse/field hockey captions on changeable panels
 7. Baseball captions on changeable panels
 8. LED Colon & Decimal
 9. Standalone Time of Day (scoreboard acts as a clock when control console is unplugged/off)
 10. Horn
 11. Individual digit protective screens

2.04 SCORING CONSOLE

- A. Console is an All Sport® 5000 controller
- B. Scores multiple sports using changeable keyboard inserts
- C. Controls multiple scoreboards and displays, including other All Sport 5000 controlled displays currently owned by customer
- D. Recalls clock, score, and period information if power is lost
- E. Console capable of automatically calculating and displaying DOWN & TO GO for each play
- F. Runs Time of Day and Segment Timer modes
- G. Console includes:
 - 1. Rugged aluminum enclosure to house electronics
 - 2. Sealed membrane water-resistant keyboard
 - 3. 32-character LCD to verify entries and recall information currently displayed
 - 4. Power cord that plugs into a standard grounded outlet; 6 watts max
 - 5. Control cable to connect to the control receptacle junction box (wired system only)
 - 6. Hand-held switch for main clock start/stop and horn
 - 7. Soft-sided carrying case
- H. Accessory Equipment
 - 1. 2.4 GHz spread spectrum radio system with frequency hopping technology and 64 non-interfering channels; system includes a transmitter installed inside the console and a receiver installed inside the scoreboard(s)
 - 2. Hard carrying case
 - 3. Battery pack

PART 3 EXECUTION

3.01 EXAMINATION

- A. Verify that mounting structure is ready to receive scoreboard. Verify that placement of conduit and junction boxes are as specified and indicated in plans and shop drawings. Verify concrete has cured adequately according to specifications.

3.02 INSTALLATION

- A. All power and control cables to scoreboards and displays will be routed in conduit. Power to the scoreboards/displays as well as raceways shown on electrical plans by the Electrical Contractor. Scoreboard control wiring including conduit will be the responsibility of the contractor assigned the scoreboard equipment.
- B. Install scoreboards and exterior displays to beams in location detailed and in accordance with manufacturer's instructions. Verify unit is plumb and level.

3.03 INSTALLATION—CONTROL CENTER

- A. Provide boxes, cover plates and jacks in locations per plans.
- B. Test connect control unit to all jacks and check for proper operation of control unit, scoreboard and all features. Leave control unit in carrying case and other loose accessories with owner's designated representative.
- C. Verify earth ground does not exceed 15 ohms.

END OF SECTION

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